From Local Forgetting to Spatial Attenuation? Reference-Stable Contraction in Partial-SWAP Collision Models.

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Abstract

We study whether local forgetting in finite-dimensional quantum memory can be composed into attenuation of information across a network. We first prove a reference-stable contraction for repeated partial-SWAP collisions between a system qubit and a memory qubit that is depolarized between collisions: for two joint system–reference states with the same reference marginal, the trace distance contracts over a two-collision block by a factor at most $1 - \epsilon$, with $\epsilon = (1 - \lambda)^2 \sin^2(2\theta)$, uniformly in the reference system and its dimension. This estimate cannot simply be iterated across a branching structure, because the reference marginals required by the hypothesis are not preserved after the first stage. We therefore analyse one explicit branching channel ($\theta = \pi/8$, $\lambda = 4/5$) through its exact Pauli transfer tensor and derive exponential attenuation bounds with rate at most about 0.73 per generation. These bounds are conditional on the recorded transfer tensor and a depth-three base case, which are numerically verified but not yet rigorously certified. We also prove, unconditionally, a single-path lower bound showing that the contraction coefficient of the depth-r channel is at least g^r with $g \approx 0.68$. The channel is nearly single-path, so it only weakly tests composition, and we identify the missing ingredients for a discriminating test: symmetric children and reuse of memory. A final section reports finite-chain numerical diagnostics and a conjectured probe-mean formula for a reused-memory model; no theorem is claimed there. No Lieb–Robinson-type or spatial attenuation bound is established.

1 Introduction and Research Problem

The propagation of correlations and information through spatially extended quantum systems is a central problem in the study of quantum many-body dynamics. Lieb–Robinson bounds provide a fundamental framework for understanding this propagation: under suitable locality assumptions, they establish that the influence of a local perturbation on a distant observable is strongly suppressed outside an effective causal region. Such bounds connect microscopic interaction structure with emergent restrictions on the speed of information propagation.

A complementary question arises when the local degrees of freedom of a quantum system interact not only with one another but also with finite-dimensional memory systems. These memories can retain information from previous interactions and thereby mediate correlations across successive stages of the dynamics. The resulting evolution is generally history-dependent when described only in terms of the system degrees of freedom. This raises a natural question: can the loss of information into, or through, local memory dynamics impose additional restrictions on the propagation of correlations beyond those arising from spatial locality alone?

A useful starting point is the notion of *reference-stable forgetting*. Informally, a local process exhibits reference-stable forgetting when it suppresses distinguishability between different inputs

even when those inputs may be correlated with an arbitrary external reference system. This qualification is important: contraction of the reduced state of a subsystem does not, by itself, establish that information correlated with an external system has also been erased. A reference-stable estimate provides a stronger local statement and is therefore a natural candidate for studying the composition of information loss across interacting subsystems.

The central research question of this work is:

> *Can local reference-stable forgetting be composed into a general spatial attenuation bound for coupled system–memory networks, potentially yielding a Lieb–Robinson-type bound beyond ordinary locality constraints?*

Answering this question requires separating two logically distinct tasks. First, one must identify and establish a local dynamical mechanism that suppresses information while remaining stable in the presence of arbitrary reference systems. Second, one must determine whether such local suppression can be composed across a network of interacting subsystems and shared or evolving memories to produce attenuation with spatial separation or network depth. The second task is not automatic: correlations generated during one stage can become part of the effective reference for subsequent stages, and the hypotheses required for a local contraction estimate need not remain satisfied after composition.

We investigate these tasks through a two-stage strategy.

Stage I: Local reference-stable contraction. We first establish a local contraction result for a specified repeated-collision model. A system qubit interacts sequentially with a memory qubit through partial-SWAP collisions, while the memory undergoes depolarization between collisions. The resulting estimate bounds the distinguishability of two joint system–reference states with the same reference marginal, uniformly over the choice and dimension of the reference. This provides a concrete and independently established example of reference-stable forgetting in a finite-dimensional memory model. The result is local: it does not, on its own, imply a spatial Lieb–Robinson bound.

Stage II: Composition under branching dynamics. We next examine whether a related attenuation mechanism can be established in a branching-channel model. In this setting, a parent system interacts sequentially with child systems, and the resulting channel is iterated across successive branching generations. We formulate the corresponding transfer-tensor description and investigate conditions under which its action contracts with branching depth. This second analysis is model-specific and logically separate from the first. In particular, the reference-stable contraction established in Stage I cannot simply be iterated: after an initial branching stage, the effective reference for a subsequent stage includes previously generated child systems, and the equal-reference-marginal condition required by the local estimate is not automatically preserved.

The two stages therefore have different roles. Stage I establishes a local dynamical mechanism under explicit assumptions. Stage II tests a possible route from local forgetting to attenuation under composition, while making the additional structural and quantitative requirements visible. Any extension to general spatially coupled system–memory networks must address these requirements rather than infer a network-level bound from a local contraction estimate alone.

Summary of results and status. Theorem I is a rigorous local statement for the stated collision model. Theorem II (Section 3.10) concerns one explicit branching channel ($\theta = \pi/8$, $\lambda = 4/5$) and is conditional on the verification items listed there. A single-path lower bound (Proposition 3) shows that the contraction coefficient of $\mathcal{E}_r$ is at least $g_\perp^r$ with $g_\perp \approx 0.681$, and the channel is

nearly single-path (the second child receives about 0.048 of the transverse influence against 0.681 for the first), so it only weakly tests composition. Section 4 reports numerical diagnostics and a conjectured formula for a reused-memory model; it contains no theorem. No Lieb–Robinson-type bound is claimed.

The purpose of this work is thus both constructive and diagnostic. It identifies a rigorous local mechanism, develops a specific branching model in which its compositional prospects can be examined, and isolates the mathematical obstacles that must be overcome before a general memory-dependent spatial attenuation theorem can be claimed. The broader Lieb–Robinson question remains the motivating research objective.

Theorem 1 (Reference-stable contraction under repeated partial-SWAP collisions)**.** *Let S and M be qubits, and let*

$$U_\theta = \exp[-i\theta\,(X_S X_M + Y_S Y_M + Z_S Z_M)]\,, \qquad \theta \in \mathbb{R}.$$

After each collision, apply the depolarizing channel to the memory,

$$\mathcal{D}_\lambda(X) = \lambda X + (1-\lambda)\,\mathrm{Tr}(X)\frac{I_M}{2}, \qquad 0 \le \lambda < 1.$$

Define the joint channel

$$\Phi_{\theta,\lambda} = (\mathrm{id}_S \otimes \mathcal{D}_\lambda) \circ \mathrm{Ad}_{U_\theta}, \qquad \mathrm{Ad}_{U_\theta}(X) = U_\theta X U_\theta^\dagger.$$

Set

$$\varepsilon = (1-\lambda)^2 \sin^2(2\theta).$$

If $\sin(2\theta) \neq 0$*, then* $\varepsilon \in (0,1]$ *and, for every finite-dimensional reference system R and all density operators ρ_{RSM} and ω_{RSM} satisfying*

$$\rho_R = \omega_R,$$

the two-collision channel obeys

$$\left\|(\mathrm{id}_R \otimes \Phi_{\theta,\lambda}^2)(\rho_{RSM} - \omega_{RSM})\right\|_1 \le (1-\varepsilon)\|\rho_{RSM} - \omega_{RSM}\|_1.$$

Consequently, for every integer $r \ge 1$,

$$\left\|(\mathrm{id}_R \otimes \Phi_{\theta,\lambda}^{2r})(\rho_{RSM} - \omega_{RSM})\right\|_1 \le (1-\varepsilon)^r\|\rho_{RSM} - \omega_{RSM}\|_1.$$

Thus the channel contracts differences between states with the same reference marginal exponentially in the number of two-collision blocks. This is a statement about the specified joint channel; it does not by itself establish a spatial Lieb–Robinson bound.

Proof. Write

$$c = \cos(2\theta), \qquad s = \sin(2\theta).$$

Up to an irrelevant global phase,

$$U_\theta = cI_{SM} - is\,\mathrm{SWAP}_{SM}.$$

Let $\{\sigma_a\}_{a=0}^3 = \{I, X, Y, Z\}$ and define

$$p_0 = \frac{1+3\lambda}{4}, \qquad p_1 = p_2 = p_3 = \frac{1-\lambda}{4}.$$

The memory depolarizing channel has Kraus operators $\{\sqrt{p_a}\sigma_a\}_{a=0}^3$. Hence a Kraus representation of $\Phi_{\theta,\lambda}^2$ is

$$L_{ba} = \sqrt{p_b p_a}\, K_{ba}, \qquad K_{ba} = (I_S \otimes \sigma_b) U_\theta (I_S \otimes \sigma_a) U_\theta, \quad a, b \in \{0,1,2,3\}.$$

Use the unnormalized Choi convention

$$J(\mathcal{T}) = \sum_{i,j} |i\rangle\langle j| \otimes \mathcal{T}(|i\rangle\langle j|).$$

Then

$$J(\Phi_{\theta,\lambda}^2) = \sum_{a,b} p_a p_b |K_{ba}\rangle\!\rangle\langle\!\langle K_{ba}|,$$

For a Kraus operator $K : \mathcal{H}_{\text{in}} \to \mathcal{H}_{\text{out}}$, define

$$|K\rangle\!\rangle = \sum_j |j\rangle_{\text{in}} \otimes K|j\rangle_{\text{in}},$$

where $\{|j\rangle_{\text{in}}\}$ is an orthonormal basis of the input space. Since $p_a \geq p = (1-\lambda)/4$ for every a,

$$J(\Phi_{\theta,\lambda}^2) \succeq p^2 J(\Lambda), \qquad \Lambda(X) = \sum_{a,b} K_{ba} X K_{ba}^\dagger.$$

We compute the Choi spectrum of Λ. The Pauli-twirl identity

$$\sum_{a=0}^{3} (I \otimes \sigma_a) Y (I \otimes \sigma_a) = 2\ \mathrm{Tr}_M(Y) \otimes I_M$$

applied to the inner and outer sums gives

$$\Lambda(X) = 4\, F(E(X)) \otimes I_M, \qquad E(X) = \mathrm{Tr}_M(U_\theta X U_\theta^\dagger), \quad F(\sigma) = \mathrm{Tr}_M\left[U_\theta(\sigma \otimes I_M)U_\theta^\dagger\right].$$

Expanding $U_\theta = cI - is\,\mathrm{SWAP}$ gives

$$U_\theta(\sigma \otimes I)U_\theta^\dagger = c^2\,\sigma \otimes I + s^2\, I \otimes \sigma + ics\,[\sigma \otimes I, \mathrm{SWAP}].$$

Since $\mathrm{Tr}_M[(\sigma\otimes I)\mathrm{SWAP}] = \mathrm{Tr}_M[\mathrm{SWAP}(\sigma\otimes I)] = \sigma$, the commutator term vanishes after the partial trace, and $F(\sigma) = 2c^2\sigma + s^2\,\mathrm{Tr}(\sigma) I_S$. Because $\mathrm{Tr}\, E(X) = \mathrm{Tr}\, X$,

$$\Lambda(X) = \left[8c^2 E(X) + 4s^2\,\mathrm{Tr}(X) I_S\right] \otimes I_M.$$

The map $X \mapsto \mathrm{Tr}(X)\, I_{SM}$ has Choi operator I_{16}, so

$$J(\Lambda) = 4s^2 I_{16} + 8c^2\, J(E) \otimes I_M \qquad \text{(up to a reordering of tensor factors)}.$$

The map E has the two Kraus operators $K_m = \langle m|_M U_\theta$, $m \in \{0,1\}$ (each 2×4). They satisfy $\mathrm{Tr}\, K_m^\dagger K_m = \mathrm{Tr}[U_\theta^\dagger(I_S \otimes |m\rangle\langle m|)U_\theta] = 2$ and $\mathrm{Tr}\, K_0^\dagger K_1 = \mathrm{Tr}[I_S \otimes |0\rangle\langle 1|] = 0$. Hence $J(E) = \sum_m |K_m\rangle\!\rangle\langle\!\langle K_m|$ has rank two with both nonzero eigenvalues equal to 2, and $J(E) \otimes I_M$ has eigenvalue 2 with multiplicity 4 and eigenvalue 0 with multiplicity 12. Therefore

$$\mathrm{spec}\, J(\Lambda) = \begin{cases} 4s^2, & \text{multiplicity 12,} \\ 4s^2 + 16c^2 = 4 + 12c^2, & \text{multiplicity 4.} \end{cases}$$

Since $4 + 12c^2 \geq 4s^2$, in particular $J(\Lambda) \succeq 4s^2 I_{16}$. Therefore,

$$\begin{aligned} J(\Phi_{\theta,\lambda}^2) &\succeq 4p^2 s^2 I_{16} \\ &= \frac{(1-\lambda)^2 s^2}{4} I_{16} \\ &= \varepsilon J(\mathcal{R}), \end{aligned}$$

where

$$\mathcal{R}(X) = \mathrm{Tr}(X)\frac{I_{SM}}{4}, \qquad J(\mathcal{R}) = \frac{I_{16}}{4}.$$

Here $\mathcal{R}$ denotes the completely depolarizing channel on SM; it is not to be confused with the reference system R. It follows that

$$\Phi_{\theta,\lambda}^2 = \varepsilon\mathcal{R} + (1-\varepsilon)\Psi$$

for a completely positive, trace-preserving map Ψ, whenever $\varepsilon < 1$. If $\varepsilon = 1$, the same Choi inequality and trace preservation imply $\Phi_{\theta,\lambda}^2 = \mathcal{R}$, and the conclusion follows directly.

Now let $\varepsilon < 1$, and $\Delta = \rho_{RSM} - \omega_{RSM}$. Since $\rho_R = \omega_R$,

$$\mathrm{Tr}_{SM}\,\Delta = 0, \qquad (\mathrm{id}_R \otimes \mathcal{R})(\Delta) = 0.$$

Consequently,

$$(\mathrm{id}_R \otimes \Phi_{\theta,\lambda}^2)(\Delta) = (1-\varepsilon)(\mathrm{id}_R \otimes \Psi)(\Delta).$$

Trace-norm contractivity of completely positive, trace-preserving maps on Hermitian operators gives

$$\left\|(\mathrm{id}_R \otimes \Phi_{\theta,\lambda}^2)(\Delta)\right\|_1 \leq (1-\varepsilon)\|\Delta\|_1.$$

Since the reference marginal is unchanged by $\mathrm{id}_R \otimes \Phi_{\theta,\lambda}^2$, the same argument applies after each two-collision block. Iterating the one-block inequality r times gives

$$\left\|(\mathrm{id}_R \otimes \Phi_{\theta,\lambda}^{2r})(\Delta)\right\|_1 \leq (1-\varepsilon)^r\|\Delta\|_1,$$

as claimed. If $\varepsilon = 1$, then $\Phi_{\theta,\lambda}^2 = \mathcal{R}$, so $\Phi_{\theta,\lambda}^{2r} = \mathcal{R}$ for every $r \geq 1$. Since $(\mathrm{id}_R \otimes \mathcal{R})(\Delta) = 0$, the left-hand side vanishes and both inequalities hold trivially. □

Remark 1 (Scope and tightness)**.** *The constant $\varepsilon = (1-\lambda)^2 \sin^2(2\theta)$ is a sufficient minorization constant obtained by retaining the smallest Pauli-Kraus weight. It is not asserted to be optimal: the optimal constant for minorization by $\mathcal{R}$ is $4\lambda_{\min}(J(\Phi_{\theta,\lambda}^2))$, which requires the spectrum of the* weighted *Choi operator. The calculation above uses the partial-SWAP structure explicitly; extensions to other collision unitaries require a corresponding positive Choi minorization and cannot be inferred from this proof alone.*

Relation to prior work

The contraction mechanism used in Theorem 1 is completely positive (CP-order) minorization, the mechanism underlying the quantum Doeblin coefficient. If $\Phi - \varepsilon\mathcal{R}_\tau$ is completely positive for some state τ and $\varepsilon \in (0,1)$, then the trace-distance contraction coefficient of Φ is at most $1-\varepsilon$ [1]. The Doeblin coefficient is computable as a semidefinite program and has been evaluated for depolarizing, bit-flip, and generalized amplitude-damping channels [1]. It is also submultiplicative under concatenation, $1 - \alpha(\mathcal{N} \circ \mathcal{M}) \leq (1-\alpha(\mathcal{N}))(1-\alpha(\mathcal{M}))$ [1], of which the iteration to $2r$

collisions in our theorem is an instance. Contractivity bounds for quantum channels have also been obtained from Hilbert's projective metric and the Birkhoff–Hopf theorem [2]. We use the Choi-dominance argument only as a framework; it is not claimed as new.

The contribution of this section is the explicit evaluation of that framework for the depolarized partial-SWAP collision channel $\Phi_{\theta,\lambda}$. Specifically, we obtain a closed-form spectrum of the operator $\sum_{a,b} |K_{ba}\rangle\rangle\langle\langle K_{ba}|$, namely $4\sin^2(2\theta)$ with multiplicity 12 and $4+12\cos^2(2\theta)$ with multiplicity 4, and from it the explicit minorization constant $\varepsilon = (1-\lambda)^2 \sin^2(2\theta)$.

Literature examined. To assess novelty we examined three bodies of literature.

(i) Quantum Doeblin and minorization. We examined the quantum Doeblin coefficient literature, in particular [1], which establishes the minorization-to-contraction relation and treats several example channels. We found no closed-form Choi spectrum or minorization constant for a depolarized partial-SWAP channel.

(ii) Hilbert-metric contraction. We examined Hilbert's projective metric in quantum information [2] and the Dobrushin ergodicity coefficient for Markov operators on cones [3]. These give contraction ratios in terms of the projective diameter and contain no collision-channel computation of the present kind.

(iii) Collision models and partial-SWAP channels. We surveyed the collision-model literature, including the review [4], the original homogenization and thermalization results [5, 6], the non-Markovian collision-model framework with inter-ancilla partial-swap collisions [7, 8], and recent work on incoherent collision models [9]. We also examined treatments of partial-SWAP channels in other settings: Pauli-transfer-matrix descriptions of SWAP^α channels [10] and partial SWAP with reset as an amplitude-damping channel [11]. These analyze homogenization, convergence, memory effects, or channel quantities for the bare partial SWAP, and none gives a Choi spectrum or minorization constant for the two-step channel with memory depolarization considered here.

Novelty statement. To our knowledge, a closed-form Choi spectrum for the depolarized partial-SWAP channel and the resulting explicit minorization constant have not appeared in the quantum-Doeblin or collision-model literature.

2 From Local Forgetting to Branching Attenuation

The preceding result establishes a reference-stable contraction mechanism for a specified sequence of partial-SWAP collisions. Its significance is not limited to the decay of distinguishability on a single system: because the estimate is stable under adjoining an arbitrary reference, it controls information that may be correlated with degrees of freedom external to the local system. This makes reference-stable forgetting a natural candidate for a compositional principle in spatially extended quantum dynamics.

The key distinction, however, is between a *local contraction* and a *spatial attenuation bound*. The result of the preceding section concerns repeated interactions between a system and a particular memory, under explicit dynamical assumptions. A spatial bound would require that analogous suppression persist as local processes are coupled across a network. In such a network, the memory or child systems generated at one stage may participate in subsequent stages, and correlations created during the evolution can alter the effective reference against which later contraction must be assessed.

2.1 The composability question

Let Φ denote a local channel acting on a system S, with an external reference R. A reference-stable contraction estimate has the schematic form

$$\|(\Phi \otimes \mathrm{id}_R)(\rho_{SR}) - (\Phi \otimes \mathrm{id}_R)(\sigma_{SR})\|_1 \leq \eta \, \|\rho_{SR} - \sigma_{SR}\|_1 , \qquad 0 \leq \eta < 1, \tag{1}$$

for states satisfying the prescribed reference-marginal condition. Such an estimate is powerful precisely because it does not require the system to be initially uncorrelated with R.

To obtain attenuation over multiple stages, one would like to apply a contraction estimate successively, leading schematically to a factor q^r after r stages. This reasoning is valid only if the hypotheses of the estimate remain satisfied at each stage. In particular, a reference-marginal constraint imposed at the input of one local process must be checked again for the effective reference at the next stage. It cannot be assumed to persist merely because the first stage was contractive.

This issue becomes explicit in a branching process. Suppose a parent system P produces child systems C_1 and C_2 through a channel

$$\mathcal{C} : \mathcal{B}(\mathcal{H}_P) \longrightarrow \mathcal{B}(\mathcal{H}_{C_1} \otimes \mathcal{H}_{C_2}).$$

After one branching stage, a subsequent stage acts on child systems that may be correlated with one another and with an external reference R. For the next local application, the effective reference can therefore include both R and previously generated child degrees of freedom. Even if two input states initially have the same marginal on R, their marginals on the enlarged reference $R \otimes C_1$ need not agree after the first stage.

Consequently, the reference-stable contraction established for the repeated-collision model cannot be iterated directly to infer attenuation with branching depth. The obstruction is not a failure of the local result; it is a mismatch between the hypotheses of that result and the correlated inputs generated by network composition.

2.2 A model-specific test of composition

The next section investigates this issue in a concrete branching-channel model. A parent qubit collides twice with a first child qubit and then twice with a second child qubit, each collision being a partial-SWAP interaction followed by depolarization of the child involved; both children start in $|0\rangle$. The parent is then traced out, defining a channel from the parent to the pair of children. Iterating this channel generates a branching structure.

The analysis uses a transfer-tensor representation to describe how operator components propagate through the branching channel. The objective is to determine whether suitable trace-norm estimates can be established for the resulting recursive dynamics. Such estimates would provide a model-specific route to attenuation with branching depth, but they require their own derivation and verification; they do not follow solely from the local contraction theorem.

The branching model is therefore a test of the proposed compositional mechanism rather than a general consequence of the preceding result. Its analysis also helps identify which additional properties may be necessary for composition, such as control of correlations among child systems, invariance of the relevant constrained operator space, or a contraction estimate that remains valid for the enlarged reference generated at each stage.

2.3 Scope of the transition

The progression from local forgetting to spatial attenuation involves three distinct levels of statement:

1. a local reference-stable contraction estimate for a specified dynamical process;

2. a model-specific recursive attenuation result, established under the hypotheses of the branching-channel construction;

3. a general spatial attenuation principle for coupled system–memory networks.

The first level is established in the preceding section. The second is investigated in the following section, with its verification status and assumptions stated explicitly. The third remains the broader research objective.

In particular, neither a finite-dimensional memory nor local reference-stable contraction alone is sufficient to establish a Lieb–Robinson-type bound for a general network. A successful generalization must explain how local contraction survives the accumulation and reuse of correlations, and how the resulting estimates depend on spatial separation, network structure, or propagation depth. The branching-channel analysis provides a controlled setting in which to examine these requirements.

3 Branching-Channel Attenuation: A Model-Specific Analysis

3.1 Research question, scope, and relation to Result I

Result I establishes reference-stable contraction for a specified sequential partial-SWAP collision process. Result II investigates a distinct question: whether local forgetting can compose under branching dynamics to produce attenuation of input influence with branching depth.

The central research question is:

> Under what physically meaningful assumptions can local, reference-stable forgetting be composed into a spatial attenuation principle for coupled quantum dynamics with finite-dimensional memory?

The reason Result I cannot simply be iterated is explained in Section 2.

Result II studies a particular branching channel as a tractable test model for this question. Its transfer-tensor formulation is a model-specific extension, not a corollary of Result I. The partial-SWAP mechanism shared by the two models motivates the investigation but is not a proof dependency.

The intended mathematical strategy is to:

1. specify the branching channel and its exact transfer tensor;

2. derive recursive operator identities and trace-norm inequalities;

3. establish a finite-depth base case rigorously;

4. use induction to obtain an exponential attenuation estimate;

5. determine which parts of the argument extend beyond the specified model.

The transfer-tensor representation and recursive estimates provide a concrete route toward this objective. The verification status of each ingredient is stated where it is used and summarized in Section 3.11; the conclusion of Theorem II is conditional as stated in Section 3.10.

3.2 Collision model and branching recursion

Let a parent qubit interact with two child qubits initially prepared in the state $|00\rangle$. The parent collides twice with the first child and then twice with the second child. Each collision is a partial-SWAP interaction with angle

$$\theta = \frac{\pi}{8},$$

followed by depolarization of the involved child with parameter

$$\lambda = \frac{4}{5}.$$

Explicitly, a collision between the parent P and a child C is the unitary $U_\theta = \exp[-i\theta(X_P X_C + Y_P Y_C + Z_P Z_C)]$, and the depolarization of the child is $\mathcal{D}_\lambda(\rho) = \lambda\rho + (1-\lambda)\,\mathrm{Tr}_C(\rho) \otimes I_C/2$. After the four collisions, the parent is traced out. Denote the resulting channel by

$$\mathcal{C} : \mathcal{B}(\mathbb{C}^2) \longrightarrow \mathcal{B}(\mathbb{C}^2 \otimes \mathbb{C}^2).$$

Define the recursively branched channels by

$$\mathcal{E}_0 = \mathrm{id}, \qquad \mathcal{E}_{r+1} = (\mathcal{E}_r \otimes \mathcal{E}_r) \circ \mathcal{C}.$$

Thus $\mathcal{E}_r$ maps one input qubit to 2^r output qubits.

This recursion specifies the branching model analyzed in this section. Any extension to a different collision ordering, memory architecture, or noise model requires its own derivation.

3.3 Exact Pauli transfer tensor

Use the Pauli convention

$$\sigma_0 = I, \qquad \sigma_1 = X, \qquad \sigma_2 = Y, \qquad \sigma_3 = Z.$$

Define

$$c_{ab}^{(i)} = \frac{1}{4}\,\mathrm{Tr}[(\sigma_a \otimes \sigma_b)\mathcal{C}(\sigma_i)]\,, \qquad a, b, i \in \{0, 1, 2, 3\}.$$

The corresponding Pauli expansion is

$$\mathcal{C}(\sigma_i) = \sum_{a,b=0}^{3} c_{ab}^{(i)}\, \sigma_a \otimes \sigma_b.$$

The recorded transfer tensor for the specified collision model has the following nonzero coefficients. All unlisted entries are zero.

i	(a,b)	$c_{ab}^{(i)}$
0	$(0,0)$	$1/2$
0	$(0,3)$	$309/1000$
0	$(1,1)$	$1/1250$
0	$(1,2)$	$-17/1250$
0	$(2,1)$	$17/1250$
0	$(2,2)$	$1/1250$
0	$(3,0)$	$1/50$
0	$(3,3)$	$1/1250$
1	$(0,1)$	$2/125$
1	$(0,2)$	$9/500$
1	$(1,0)$	$17/50$
1	$(1,3)$	$153/625$
1	$(2,0)$	$1/50$
1	$(2,3)$	$-8/625$
1	$(3,1)$	$-9/625$
1	$(3,2)$	$8/625$
2	$(0,1)$	$-9/500$
2	$(0,2)$	$2/125$
2	$(1,0)$	$-1/50$
2	$(1,3)$	$8/625$
2	$(2,0)$	$17/50$
2	$(2,3)$	$153/625$
2	$(3,1)$	$-8/625$
2	$(3,2)$	$-9/625$
3	$(0,3)$	$17/1000$
3	$(1,1)$	$1/1250$
3	$(1,2)$	$-17/1250$
3	$(2,1)$	$17/1250$
3	$(2,2)$	$1/1250$
3	$(3,0)$	$17/50$
3	$(3,3)$	$289/1250$

The recorded tensor satisfies the trace-consistency identities

$$\operatorname{Tr}\mathcal{C}(I) = 2, \qquad \operatorname{Tr}\mathcal{C}(X) = \operatorname{Tr}\mathcal{C}(Y) = \operatorname{Tr}\mathcal{C}(Z) = 0.$$

Verification status. The 31 listed nonzero coefficients have been compared with a direct numerical simulation of the stated circuit and agree to machine precision. This comparison is a numerical cross-check; it is not an exact-arithmetic derivation of the transfer tensor. The coefficient sums used in (II.2)–(II.3) have been expanded term by term from the recorded entries. The depth-three norms were evaluated in double precision ($x_3 \approx 0.32020$, $z_3 \approx 0.31535$), below the proposed threshold $1/3$. These values remain numerical diagnostics.

An exact-arithmetic construction of the transfer tensor and of $A_i^{(3)}$, together with a rigorous trace-norm certificate, has not yet been recorded. A numerical spectrum with an error matrix E can be used only after the construction and the applicable error bound have been specified and verified.

3.4 Coefficient absolute sums

Define

$$L_i = \sum_{a,b=0}^{3} |c_{ab}^{(i)}|.$$

Summing the absolute values of the recorded coefficients gives

$$L_0 = \frac{4293}{5000} = 0.8586,$$

$$L_1 = L_2 = \frac{1697}{2500} = 0.6788, \qquad L_3 = \frac{617}{1000} = 0.617.$$

Consequently,

$$\max_i L_i = \frac{4293}{5000} < 1.$$

These are coefficient absolute sums for the one-step transfer tensor. They are not, by themselves, a proof of trace-norm contraction for the recursively branched channel. In particular, the inequality $\max_i L_i < 1$ cannot alone establish the desired reference-stable attenuation estimate.

3.5 Exact recursive operator representation

For $i \in \{0, 1, 2, 3\}$, define

$$A_i^{(r)} = \mathcal{E}_r(\sigma_i).$$

The initial conditions are

$$A_i^{(0)} = \sigma_i,$$

and the transfer-tensor expansion yields the exact recurrence

$$\boxed{A_i^{(r+1)} = \sum_{a,b=0}^{3} c_{ab}^{(i)} A_a^{(r)} \otimes A_b^{(r)}.} \tag{II.1}$$

Since each $\mathcal{E}_r$ is completely positive and trace-preserving,

$$A_0^{(r)} = \mathcal{E}_r(I) \succeq 0, \qquad \operatorname{Tr} A_0^{(r)} = 2.$$

For $i = 1, 2, 3$, the operators $A_i^{(r)}$ are Hermitian and traceless.

Equation (II.1) provides a finite recursive construction of the operators at every depth. The remaining challenge is to control their trace norms with estimates that remain useful under repeated branching.

3.6 Recursive trace-norm estimates

Define

$$x_r = \max\left\{\frac{1}{2}\|A_1^{(r)}\|_1, \frac{1}{2}\|A_2^{(r)}\|_1\right\}, \qquad z_r = \frac{1}{2}\|A_3^{(r)}\|_1.$$

At depth zero,

$$x_0 = z_0 = 1.$$

Using the transfer-tensor recurrence, the triangle inequality, and the tensor-product identity

$$\|B \otimes D\|_1 = \|B\|_1 \|D\|_1,$$

the proposed recursive estimates are

$$\boxed{x_{r+1} \le (0.788 + 0.5696 z_r) x_r,} \tag{II.2}$$

and

$$\boxed{z_{r+1} \le 0.714 z_r + 0.4624 z_r^2 + 0.0576 x_r^2.} \tag{II.3}$$

The finite decimals above are exact rationals: $0.788 = \frac{197}{250}$, $0.5696 = \frac{356}{625}$, $0.714 = \frac{357}{500}$, $0.4624 = \frac{289}{625}$, $0.0576 = \frac{36}{625}$.

Lemma 1 (Recursive inequalities)**.** *Using the entries $c_{ab}^{(i)}$ tabulated above, the operators generated by (II.1) satisfy (II.2) and (II.3) for every $r \ge 0$.*

Proof. We have $\|A_0^{(r)}\|_1 = 2$, $\|A_1^{(r)}\|_1, \|A_2^{(r)}\|_1 \le 2x_r$ and $\|A_3^{(r)}\|_1 = 2z_r$. Apply the triangle inequality and $\|B \otimes D\|_1 = \|B\|_1 \|D\|_1$ to (II.1). For $i = 1$: the entries with $(a, b) \in \{(1, 0), (2, 0), (0, 1), (0, 2)\}$ have absolute sum $\frac{17}{50} + \frac{1}{50} + \frac{2}{125} + \frac{9}{500} = \frac{197}{500}$ and multiply operators of trace norm at most $4x_r$; the entries with $(a, b) \in \{(1, 3), (2, 3), (3, 1), (3, 2)\}$ have absolute sum $\frac{153+8+9+8}{625} = \frac{178}{625}$ and multiply operators of trace norm at most $4x_r z_r$. Hence $\frac{1}{2}\|A_1^{(r+1)}\|_1 \le \frac{197}{250} x_r + \frac{356}{625} x_r z_r$. The entries for $i = 2$ have the same two absolute sums, which gives (II.2). For $i = 3$: the entries $(0, 3), (3, 0)$ have absolute sum $\frac{357}{1000}$ and multiply operators of norm at most $4z_r$; the entries $(a, b) \in \{1, 2\}^2$ have sum $\frac{36}{1250}$ and norm at most $4x_r^2$; the entry $(3, 3)$ is $\frac{289}{1250}$ with norm $4z_r^2$. Hence $z_{r+1} \le \frac{357}{500} z_r + \frac{36}{625} x_r^2 + \frac{289}{625} z_r^2$, which is (II.3). □

The rotation-aware sharpening of (II.2) is given in Proposition 2.

Lemma 2 ($U(1)$ covariance)**.** *Let $W_\varphi = e^{-i\varphi Z/2}$ and let $N = 2^r$ be the number of output qubits of $\mathcal{E}_r$. For every $r \ge 0$ and every $\varphi \in \mathbb{R}$,*

$$\mathcal{E}_r(W_\varphi \cdot W_\varphi^\dagger) = \mathcal{W}_\varphi^{(r)} \circ \mathcal{E}_r, \qquad \mathcal{W}_\varphi^{(r)}(\,\cdot\,) = W_\varphi^{\otimes N}(\,\cdot\,) W_\varphi^{\dagger \otimes N}.$$

Proof. Since $X_S X_M + Y_S Y_M + Z_S Z_M = 2\,\mathrm{SWAP} - I$ and SWAP commutes with $V \otimes V$ for every unitary V, we have $[U_\theta, W_\varphi \otimes W_\varphi] = 0$. The depolarizing channel satisfies $\mathcal{D}_\lambda(WXW^\dagger) = W \mathcal{D}_\lambda(X) W^\dagger$, because Tr and I are invariant under conjugation. Finally $W_\varphi |0\rangle \propto |0\rangle$, so $|00\rangle\langle 00|$ is invariant. Hence, for the parent P and children C_1, C_2,

$$W_\varphi \rho W_\varphi^\dagger \otimes |00\rangle\langle 00| = (W_\varphi^{\otimes 3})(\rho \otimes |00\rangle\langle 00|)(W_\varphi^{\otimes 3})^\dagger.$$

Each collision and depolarization commutes with $W_\varphi^{\otimes 3}$. Tracing out P removes its factor, so $\mathcal{C}(W_\varphi \rho W_\varphi^\dagger) = (W_\varphi \otimes W_\varphi)\, \mathcal{C}(\rho)\, (W_\varphi \otimes W_\varphi)^\dagger$. For $r + 1$, $\mathcal{E}_{r+1} = (\mathcal{E}_r \otimes \mathcal{E}_r) \circ \mathcal{C}$ and the induction hypothesis give the claim. □

Lemma 3 (Rotation identity)**.** *For all $r \ge 0$ and all real u, v,*

$$\|u A_1^{(r)} + v A_2^{(r)}\|_1 = \sqrt{u^2 + v^2}\, \|A_1^{(r)}\|_1, \qquad \|A_2^{(r)}\|_1 = \|A_1^{(r)}\|_1.$$

In particular $x_r = \frac{1}{2}\|A_1^{(r)}\|_1 = \frac{1}{2}\|A_2^{(r)}\|_1$.

Proof. Since $W_\varphi X W_\varphi^\dagger = \cos\varphi\, X + \sin\varphi\, Y$, linearity and Lemma 2 give $\cos\varphi\, A_1^{(r)} + \sin\varphi\, A_2^{(r)} = \mathcal{W}_\varphi^{(r)}(A_1^{(r)})$. Choose φ with $(\cos\varphi, \sin\varphi) = (u, v)/\sqrt{u^2 + v^2}$ (the case $u = v = 0$ is trivial). The trace norm is invariant under unitary conjugation. Taking $\varphi = \pi/2$ gives the second claim. □

3.7 Depth-three base case

The proposed induction requires the finite-depth estimate

$$x_3 \le \frac{1}{3}, \qquad z_3 \le \frac{1}{3}, \tag{II.4}$$

equivalently,

$$\|A_1^{(3)}\|_1 \le \frac{2}{3}, \qquad \|A_2^{(3)}\|_1 \le \frac{2}{3}, \qquad \|A_3^{(3)}\|_1 \le \frac{2}{3}. \tag{II.5}$$

Earlier floating-point calculations reported the approximate values

$$x_3 \approx 0.320196440329, \qquad z_3 \approx 0.315352936004,$$

corresponding to the estimates

$$\|A_1^{(3)}\|_1 = \|A_2^{(3)}\|_1 \approx 0.640392880658,$$

$$\|A_3^{(3)}\|_1 \approx 0.630705872008.$$

These numbers are numerical diagnostics, not rigorous certificates. Although they lie below the proposed thresholds in (II.4)–(II.5), numerical proximity to the required inequalities does not prove them.

An exact-integer certificate was previously claimed, but its computation and output have not been independently reproduced and checked. Earlier characteristic-polynomial calculations also contained errors and are not relied upon here.

Status of the base case. The construction of the depth-three operators is substantively specified by (II.1). The required trace-norm inequalities remain pending a reproducible rigorous certificate, such as exact-arithmetic bounds or independently verified eigenvalue enclosures sufficient to establish (II.5).

3.8 Conditional induction

Assume for this subsection that the recorded transfer tensor correctly represents the branching channel and that the depth-three base case (II.4) holds; (II.2)–(II.3) then hold by Lemma 1. Under these hypotheses, the following induction establishes the stated attenuation estimates.

Set

$$q = 0.788 + \frac{0.5696}{3} = \frac{3667}{3750} < 1. \tag{II.6}$$

We then obtain the candidate bounds

$$\boxed{x_{3+n} \le \frac{q^n}{3}, \qquad z_{3+n} \le \frac{q^{2n}}{3}, \qquad n \ge 0.} \tag{II.7}$$

We prove (II.7) by induction on n, carrying both bounds together; the case $n = 0$ is (II.4). Suppose $x_{3+n} \le q^n/3$ and $z_{3+n} \le q^{2n}/3$, so in particular $z_{3+n} \le \frac{1}{3}$. Then (II.2) gives

$$\begin{aligned} x_{4+n} &\le (0.788 + 0.5696 z_{3+n}) x_{3+n} \\ &\le q\,\frac{q^n}{3} \\ &= \frac{q^{n+1}}{3}. \end{aligned}$$

For the longitudinal component, $z_{3+n}^2 \le \frac{1}{3} z_{3+n} \le q^{2n}/9$ and $x_{3+n}^2 \le q^{2n}/9$, so (II.3) gives

$$\begin{aligned} z_{4+n} &\le 0.714 z_{3+n} + 0.4624 z_{3+n}^2 + 0.0576 x_{3+n}^2 \\ &\le \frac{q^{2n}}{3}\left(0.714 + \frac{0.4624}{3} + \frac{0.0576}{3}\right). \end{aligned}$$

The factor in parentheses equals

$$0.714 + \frac{0.4624 + 0.0576}{3} = \frac{1331}{1500} < q^2, \qquad q^2 = \left(\tfrac{3667}{3750}\right)^2 \approx 0.9562.$$

Therefore,

$$z_{4+n} \le \frac{q^{2(n+1)}}{3}.$$

This proves (II.7) by induction, conditional on the transfer tensor and (II.4). In particular, the induction closes within the region $x_r, z_r \le 1/3$ once the stated base case and recursive inequalities hold.

Proposition 1 (Refined rate)**.** *Assume (II.4), and put $s = 0.788$, $q = 3667/3750$. Then for all $n \ge 0$,*

$$x_{3+n} \le K\, s^n, \qquad K = \tfrac{1}{3}\exp\!\Big(\tfrac{0.5696}{3\,s\,(1-q^2)}\Big) < 82.$$

Moreover, for every $\mu \in (s, 1)$ there is $C_\mu < \infty$ with $z_{3+n} \le C_\mu \mu^n$. Hence the bound (II.8) decays as $2x_r + z_r \le C'_\mu\, \mu^{r-3}$ for every $\mu > 0.788$.

Proof. By (II.2), $x_{4+n} \le (s + 0.5696\, z_{3+n}) x_{3+n}$, so

$$x_{3+n} \le x_3 s^n \prod_{k<n}\Big(1 + \tfrac{0.5696}{s} z_{3+k}\Big) \le \tfrac{1}{3} s^n \exp\!\Big(\tfrac{0.5696}{s}\sum_{k\ge 0} z_{3+k}\Big).$$

Since $z_{3+k} \le q^{2k}/3$, the sum is at most $\frac{1}{3(1-q^2)}$, which gives K. For z, let $a_n = 0.714 + 0.4624 z_{3+n}$, so $z_{4+n} \le a_n z_{3+n} + 0.0576\, x_{3+n}^2$. Since $z_{3+n} \to 0$, there is N with $a_n \le \mu$ for $n \ge N$. Put $w_n = z_{3+n}\mu^{-n}$. Then $w_{n+1} \le w_n + h_n$ with $h_n = 0.0576 K^2 s^{2n} \mu^{-(n+1)}$, which is summable because $s^2 < \mu$. Hence w_n is bounded. □

Proposition 2 (Rotation-aware recursion and rate)**.** *Define, from the transfer tensor,*

$$\begin{aligned} g_0 &= \sqrt{(c_{10}^{(1)})^2 + (c_{20}^{(1)})^2}, \quad g_1 = \sqrt{(c_{01}^{(1)})^2 + (c_{02}^{(1)})^2}, \\ h_0 &= \sqrt{(c_{13}^{(1)})^2 + (c_{23}^{(1)})^2}, \quad h_1 = \sqrt{(c_{31}^{(1)})^2 + (c_{32}^{(1)})^2}, \end{aligned}$$

and $s' = 2(g_0 + g_1)$, $\kappa = 2(h_0 + h_1)$. For the model of Section 3.2, $g_0 = \sqrt{0.116}$, $g_1 = \sqrt{0.00058}$, $h_0 = \sqrt{0.06009088}$, $h_1 = \sqrt{0.0003712}$, hence

$$s' \le 0.7294, \qquad \kappa \le 0.5289.$$

Then for all $r \ge 0$,

$$x_{r+1} \le (s' + \kappa z_r)\, x_r. \tag{II.2$'$}$$

If, in addition, (II.4) holds, then for all $n \ge 0$,

$$x_{3+n} \le K'\, s'^n, \qquad K' = \tfrac{1}{3}\exp\!\Big(\tfrac{\kappa}{3\,s'(1-q^2)}\Big) < 84,$$

with $q = 3667/3750$ as in (II.6), and for every $\mu \in (s', 1)$ there is $C_\mu < \infty$ with $z_{3+n} \le C_\mu \mu^n$. Consequently

$$2x_r + z_r \le C'_\mu\, \mu^{r-3} \qquad \text{for every } \mu > 0.7294.$$

Proof. By (II.1) with $i = 1$, the eight nonzero terms of $A_1^{(r+1)}$ group into four pairs sharing one tensor factor:

$$A_1^{(r+1)} = (c_{10}^{(1)} A_1 + c_{20}^{(1)} A_2) \otimes A_0 + A_0 \otimes (c_{01}^{(1)} A_1 + c_{02}^{(1)} A_2) + (c_{13}^{(1)} A_1 + c_{23}^{(1)} A_2) \otimes A_3 + A_3 \otimes (c_{31}^{(1)} A_1 + c_{32}^{(1)} A_2),$$

all at depth r. Apply the triangle inequality across the four pairs, $\|B \otimes D\|_1 = \|B\|_1 \|D\|_1$, Lemma 3 within each pair, $\|A_0^{(r)}\|_1 = 2$, $\|A_1^{(r)}\|_1 = 2x_r$ and $\|A_3^{(r)}\|_1 = 2z_r$:

$$\tfrac{1}{2}\|A_1^{(r+1)}\|_1 \le \tfrac{1}{2}\big[\, 2g_0 \cdot 2x_r + 2g_1 \cdot 2x_r + h_0 \cdot 2x_r \cdot 2z_r + h_1 \cdot 2x_r \cdot 2z_r \big] = (s' + \kappa z_r) x_r.$$

By Lemma 3, $x_{r+1} = \tfrac{1}{2}\|A_1^{(r+1)}\|_1$, giving (II.2′). For the rate, $x_{r+1} \le (s' + \kappa z_r) x_r$ and the bound $z_{3+k} \le q^{2k}/3$ from the induction of Section 3.8 (still valid, since $x_{3+n} \le q^n/3$ follows from (II.2′) because $s' + \kappa/3 < q$) give

$$x_{3+n} \le \tfrac{1}{3} s'^{\,n} \exp\Big(\tfrac{\kappa}{s'} \sum_{k \ge 0} z_{3+k} \Big) \le \tfrac{1}{3} s'^{\,n} \exp\Big(\tfrac{\kappa}{3s'(1-q^2)} \Big).$$

The bound on z_{3+n} follows exactly as in Proposition 1, using $\mu > s' > 0.714$ and $s'^2 < \mu$. The last claim is (II.8). □

3.9 Reference-stable influence estimate

Let R be an arbitrary reference system and let ρ, ω be states on the input qubit and R, with equal reference marginals:

$$\mathrm{Tr}_S\, \rho = \mathrm{Tr}_S\, \omega,$$

where S is the input qubit.

Writing the input difference in the Pauli basis gives

$$\rho - \omega = \frac{1}{2} \sum_{i=1}^{3} \sigma_i \otimes D_i,$$

where

$$D_i = \mathrm{Tr}_S\big[(\sigma_i \otimes I_R)(\rho - \omega)\big].$$

Since the reference marginals agree, the identity-Pauli coefficient vanishes.

Applying $\mathcal{E}_r$ to the system yields

$$(\mathcal{E}_r \otimes \mathrm{id}_R)(\rho - \omega) = \frac{1}{2} \sum_{i=1}^{3} A_i^{(r)} \otimes D_i.$$

For each i, since $\sigma_i \otimes I_R$ is unitary and the partial trace is trace-norm contractive,

$$\|D_i\|_1 \le \|\rho - \omega\|_1.$$

Consequently, by the triangle inequality,

$$\begin{aligned} \|(\mathcal{E}_r \otimes \mathrm{id}_R)(\rho - \omega)\|_1 &\le \frac{1}{2} \sum_{i=1}^{3} \|A_i^{(r)}\|_1 \|D_i\|_1 \\ &\le \left(\frac{1}{2}\|A_1^{(r)}\|_1 + \frac{1}{2}\|A_2^{(r)}\|_1 + \frac{1}{2}\|A_3^{(r)}\|_1 \right) \|\rho - \omega\|_1 \\ &\le (2x_r + z_r)\|\rho - \omega\|_1. \end{aligned}$$

Thus the proposed reference-stable influence estimate is

$$\boxed{\|(\mathcal{E}_r \otimes \mathrm{id}_R)(\rho-\omega)\|_1 \le (2x_r + z_r)\|\rho-\omega\|_1.} \tag{II.8}$$

The ordering of tensor factors is immaterial to the trace norm; equivalently, one may write the channel action as $\mathrm{id}_R \otimes \mathcal{E}_r$ after a suitable factor permutation.

This estimate follows from the Pauli decomposition and the stated trace-norm inequalities. It does not require the reference system to be uncorrelated with the input. It does require the input states to have equal reference marginals.

3.10 Conditional branching-attenuation theorem

Theorem II (conditional, model-specific). For the branching channel defined in Section 3.2, suppose that:

1. the transfer tensor in Section 3.3 correctly represents the channel; and
2. the depth-three base-case inequalities (II.4), $x_3 \le \frac{1}{3}$ and $z_3 \le \frac{1}{3}$, hold.

Then for every $r \ge 3$, every reference system R, and all input states ρ, ω with equal reference marginals,

$$\boxed{\|(\mathcal{E}_r \otimes \mathrm{id}_R)(\rho-\omega)\|_1 \le q^{r-3}\|\rho-\omega\|_1, \qquad q = \frac{3667}{3750} < 1.} \tag{II.9}$$

Proof under the stated hypotheses. The induction in Section 3.8 gives

$$x_r \le \frac{q^{r-3}}{3}, \qquad z_r \le \frac{q^{2(r-3)}}{3}.$$

Substituting these inequalities into (II.8) yields

$$\begin{aligned}\|(\mathcal{E}_r \otimes \mathrm{id}_R)(\rho-\omega)\|_1 &\le \left(\frac{2}{3}q^{r-3} + \frac{1}{3}q^{2(r-3)}\right)\|\rho-\omega\|_1 \\ &\le q^{r-3}\|\rho-\omega\|_1,\end{aligned}$$

because $0 < q < 1$ and $r \ge 3$. □

Proof status. The implication from hypotheses (1)–(2) to (II.9) is proved by the induction and influence estimate given above. The result is not an unconditional theorem: the transfer tensor currently has a numerical cross-check rather than a recorded exact derivation, and the depth-three base case (II.4) has not been rigorously certified. These items must be resolved before Theorem II can be stated without its hypotheses.

Proposition 3 (Single-path lower bound)**.** *Let $\mathcal{R}(\cdot) = \mathrm{Tr}_{C_2}\mathcal{C}(\cdot)$ be the channel to the first child. Its Bloch action on the transverse plane is the rotation–dilation $\begin{smallmatrix}\alpha & \beta\\ -\beta & \alpha\end{smallmatrix}$ and $\sigma_3 \mapsto \alpha\sigma_3$, with*

$$\alpha = 2c_{10}^{(1)} = \lambda(1+3\lambda)\,c_\theta^2 s_\theta^2, \qquad \beta = 2c_{20}^{(1)} = \lambda\, c_\theta s_\theta\big[(1+\lambda)s_\theta^2 - 2\lambda c_\theta^2\big],$$

with $c_\theta = \cos 2\theta$, $s_\theta = \sin 2\theta$. At $\theta = \pi/8$, $\lambda = 4/5$: for all r,

$$\sup_{\rho_R=\omega_R} \frac{\|(\mathcal{E}_r \otimes \mathrm{id}_R)(\rho-\omega)\|_1}{\|\rho-\omega\|_1} \;\ge\; g_\perp^r, \qquad g_\perp = \sqrt{\alpha^2+\beta^2} \approx 0.6812.$$

Proof. Each $\mathcal{E}_{r-1}$ is trace preserving, so tracing out everything except the leftmost leaf gives $\mathcal{R}^r$ by induction. Take $\rho = |+\rangle\langle+|$, $\omega = |-\rangle\langle-|$, so $\rho - \omega = \sigma_1$ and $\|\rho - \omega\|_1 = 2$. Then $\mathcal{R}^r(\sigma_1)$ is a transverse Bloch vector of length $g_\perp^r$, with trace norm $2g_\perp^r$. The partial trace is trace-norm contractive, so the full output norm is at least $2g_\perp^r$. □

Remark 2 (Rate bracket)**.** *Let η_r be the supremum in Proposition 3. Then $\eta_r \geq g_\perp^r$ with $g_\perp \approx 0.6812$. If the transfer tensor and (II.4) hold, Proposition 2 and (II.8) give $\eta_r \leq 2x_r + z_r \leq C'_\mu \mu^{r-3}$ for every $\mu > 0.7294$. Hence, under these hypotheses, $0.6812 \leq \liminf \eta_r^{1/r} \leq \limsup \eta_r^{1/r} \leq 0.7294$.*

3.11 What the branching model does and does not establish

The analysis distinguishes four levels of conclusion.

Exact algebraic formulation. The branching channel, Pauli transfer-tensor definition, and operator recursion (II.1) specify the model and its recursive representation. The listed coefficient sums follow by summing the absolute values of the recorded tensor entries.

Analytical route. The recursive inequalities (II.2)–(II.3) and the induction identify a proposed route from finite-depth control to exponential attenuation. The induction closes conditional on the base case and the recursive inequalities.

Numerical evidence. The reported depth-three trace-norm values lie below the threshold required by the proposed induction. They are diagnostic evidence only; without a reproducible rigorous certificate, they cannot establish the base case.

Open theorem. The unconditional branching-attenuation result remains pending. The transfer tensor has been numerically cross-checked, the recursive coefficient estimates have been checked term by term, and the depth-three values have been evaluated numerically. These checks do not yet constitute a complete rigorous certificate. Equation (II.9) is therefore a conditional result, with the hypotheses stated in Theorem II. The same qualification applies to the sharper rates of Propositions 1 and 2.

3.12 Diagnostic limitation and central open problem

The branching model is intended to test whether a local reference-stable contraction mechanism can survive composition through a network. The analysis does not establish a general Lieb–Robinson bound and does not derive a spatial attenuation law from Result I alone.

In particular, finite memory dimension and mixing of a reference channel, without stronger assumptions controlling how information is transmitted and reused, do not by themselves establish a memory-induced spatial bottleneck. Local marginal convergence is not sufficient to establish reference-stable forgetting, and local forgetting does not automatically imply attenuation under arbitrary branching.

The central open problem is therefore to identify physically interpretable assumptions under which local reference-stable forgetting composes into a spatial attenuation principle for coupled system–memory dynamics. The present branching model provides a concrete test case for this compositional question, not a proof of the general principle.

Remark 3 (Weak branching and built-in geometry)**.** *The gain from the parent to the second child is* $\|(\textit{transverse gain})\| = \sqrt{0.032^2 + 0.036^2} \approx 0.048$*, against* 0.681 *to the first child. The influence is carried by one dominant path (Proposition 3); the second branch is nearly inert, so the model barely tests composition under branching. The tree is also feed-forward: children are fresh, the parent is discarded, and no memory is reused. A discriminating test requires (i) a symmetric-children ordering, and (ii) reuse of memory across generations. Both are open.*

3.13 Outstanding verification tasks

Before Result II can be presented as an unconditional branching-attenuation theorem, the following tasks remain:

1. **Derive the transfer tensor in exact arithmetic.** Reproduce the 31 nonzero coefficients from the stated collision and depolarization conventions, with an explicit and reproducible calculation.
2. **Certify the depth-three base case.** Construct $A_i^{(1)}, A_i^{(2)}, A_i^{(3)}$ from (II.1) and prove the three trace-norm bounds in (II.5), using a reproducible exact-arithmetic certificate or a rigorously enclosing numerical method.
3. **Complete an end-to-end verification.** Starting from the certified tensor, verify the recursive inequalities, the induction hypotheses, and every constant in (II.6)–(II.9).
4. **Assess generality.** Determine whether the contraction mechanism extends to less restrictive, physically meaningful coupled system–memory models without assuming away the influence carriers that the broader research question is intended to control.

Until these tasks are completed, the rigorous attenuation conclusion remains conditional on the hypotheses of Theorem II. The broader compositional principle remains an open research question.

4 Shared-Memory Extension: Transfer Representation and Finite-Chain Diagnostics

4.1 Motivation and distinction from the branching model

The branching channel in Section 3 uses fresh child systems at each generation and discards the parent after each branching step. The present extension investigates a different architecture: a single persistent memory is reused while interacting sequentially with sites of a finite physical chain. Consequently, correlations created during one part of a sweep may remain in the memory and affect later sites or subsequent sweeps.

The purpose of this investigation is to test whether local reference-stable forgetting can compose uniformly in system size when memory is reused. This is a distinct problem from the branching-channel attenuation estimate, and no result in this section is asserted to follow from Result I or Result II.

4.2 Shared-memory collision model

Consider physical qubits $S_1, \ldots, S_N$ and one persistent memory qubit M. At each site, the memory interacts with the physical qubit through the partial-SWAP unitary

$$U_{iM}(\theta) = \exp[-i\theta\,(X_iX_M + Y_iY_M + Z_iZ_M)]\,, \qquad \theta = \frac{\pi}{8},$$

up to an irrelevant global phase. After each collision, the memory is depolarized according to

$$\mathcal{D}_\lambda(X) = \lambda X + (1-\lambda)\,\mathrm{Tr}(X)\frac{I_M}{2}, \qquad \lambda = \frac{4}{5}.$$

A sweep visits the sites in the order $S_1, S_2, \ldots, S_N$. The same physical sites and the same memory are reused on subsequent sweeps. In particular, the memory output at the end of one sweep is the memory input at the beginning of the next sweep. The model therefore retains inter-sweep memory correlations; it is not a fresh-ancilla circuit.

The precise multi-sweep channel is to be defined by the chronological composition of these collisions and memory-noise maps, with the open-chain boundary wiring retained explicitly.

4.3 Finite-chain probe diagnostics

A finite-chain diagnostic was performed with $L = 2$ sweeps (sites visited in the order $S_1, \ldots, S_N, S_1, \ldots, S_N$, the memory depolarized after every collision), using the input

$$\bigotimes_{i=1}^{N} |\Phi^+\rangle\langle\Phi^+|_{S_iR_i} \otimes \frac{I_M}{2}, \qquad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

where each physical site S_i is paired with a reference qubit R_i. The corresponding comparison input replaces each $|\Phi^+\rangle\langle\Phi^+|_{S_iR_i}$ by $I_{S_iR_i}/4$, retaining the same memory marginal.

For the evolved state, define the local probe

$$P_i^\Phi = |\Phi^+\rangle\langle\Phi^+|_{S_iR_i}.$$

The mean probe value and the connected covariance are

$$m_i^{(N)} = \mathrm{Tr}\Big(P_i^\Phi\, \rho_{\mathrm{out}}^{(N)}\Big),$$

and

$$\mathrm{Cov}^{(N)}(P_i^\Phi, P_j^\Phi) = \mathrm{Tr}\Big(P_i^\Phi P_j^\Phi \rho_{\mathrm{out}}^{(N)}\Big) - m_i^{(N)} m_j^{(N)}, \qquad i \neq j.$$

Here the probe means and covariances refer to the maximally-entangled-probe input described above.

In all computed cases $m_i^{(N)}$ is independent of i, and $\mathrm{Cov}^{(N)}$ depends only on $|i-j|$; m_N denotes the common value.

Numerical diagnostic data. The following values were obtained from direct chronological finite-chain calculations. They are reported as numerical diagnostics, not as exact-arithmetic certificates.

N	single-probe mean m_N
1	0.2875
2	0.3775
3	0.4135
4	0.4279
5	0.43366

The recorded two-probe connected covariances are

N	separation $\lvert i-j\rvert$	$\mathrm{Cov}^{(N)}(P_i^\Phi, P_j^\Phi)$
2	1	0.036900
3	1	0.033624
3	2	0.025284
4	1	0.03158784
4	2	0.01889184
4	3	0.01432704

These finite-size values suggest that the probe means approach a limiting value and that the connected covariances may attenuate with separation. They do not establish either behavior uniformly in N. In particular, the data do not by themselves prove a summable covariance estimate, a collective distinguishability bound, or a uniform reference-stable contraction estimate.

4.4 Conjectured probe-mean formula

The finite-chain means above are consistent with the conjectural formula

$$\boxed{m_N = \frac{7}{16} - \frac{3}{20}\left(\frac{2}{5}\right)^{N-1}, \qquad N \geq 1.} \tag{SM.1}$$

This expression is a finite-data conjecture only. It contains three constants (7/16, 3/20 and 2/5) fitted to the data, so the values for $N = 1, \ldots, 4$ provide only one independent check; the value $m_5 = 0.43366$ is a further check that (SM.1) reproduces. It has not been derived from an exact transfer representation, and agreement at finitely many N does not establish (SM.1) for arbitrary N. It is retained as a testable hypothesis for the transfer reconstruction.

4.5 Transfer representation: unresolved reconstruction

A transfer representation may provide a tractable way to analyze the repeated-sweep circuit and its probe observables. Such a representation must retain the correlations of each physical site between its encounters and must correctly encode the memory boundary between successive sweeps.

In particular, the transfer construction must explicitly account for the connection between the end of sweep 1 and the start of sweep 2. Treating these boundaries as independent, or replacing the persistent memory by fresh ancillary systems, would describe a different channel.

The sparse 16×16 transfer and probe-insertion matrices previously discussed are not included here: their Kraus-index contraction, boundary wiring, and agreement with the direct chronological circuit have not been independently reconstructed and verified. No spectral values, eigenvectors, or probe overlaps from that unverified construction are used in this section.

Verification status. The finite-chain data in Section 4.3 are numerical diagnostics. A reproducible exact-arithmetic transfer construction remains outstanding. The transfer matrix, probe-insertion operators, and boundary vectors must be derived from the Kraus representation of the collision and depolarization maps, with a fixed matrix-unit ordering and explicit index contractions. The resulting representation must reproduce the direct chronological probe means and covariances for the finite chains already evaluated before its spectrum is used for asymptotic conclusions.

4.6 Research program: uniform contraction and collective probes

The next steps are:

1. **Derive the local transfer tensor.** Obtain the collision superoperator from its Kraus operators and specify the tensor index convention, including the physical-site degrees of freedom retained between encounters.

2. **Wire the open-chain multi-sweep boundary.** Construct the exact connection from the memory output at the end of sweep 1 to the memory input at the start of sweep 2. Keep this connection explicit in the transfer representation.

3. **Construct and validate the finite-chain transfer.** Derive the transfer matrix, probe-insertion operators, and boundary data. Reproduce the means and covariances for $N = 1, \ldots, 4$ using a reproducible calculation and compare against the direct chronological circuit.

4. **Certify the asymptotic transfer behavior.** Only after finite-chain validation, determine the exact or rigorously enclosed spectrum, identify the dominant subleading modes, and compute their overlaps with the boundary and probe-insertion operators. A spectral gap alone is not sufficient to establish a uniform estimate without controlling the associated prefactors and overlaps.

5. **Establish uniform probe bounds.** Seek a rigorous large-N estimate for the probe means and connected covariances, including a summable covariance bound uniform in chain length if the data support such a result.

6. **Test collective distinguishability.** Consider the family of inputs in which each physical site is entangled with its own reference qubit. Determine whether the collective output distinguishability can remain bounded away from zero as N grows, despite local probe attenuation. Such a result would provide a candidate obstruction to a uniform reference-stable contraction estimate.

7. **Address fixed-block contraction.** Define the reference-stable trace-norm contraction coefficient $\eta_N(L)$ for a specified L-sweep block, allowing arbitrary reference systems and input differences with equal reference marginals. The central target is to determine whether there exist a fixed L and $q < 1$ such that

$$\sup_{N \geq 1} \eta_N(L) \leq q.$$

   If this fails, seek a rigorous lower-bound construction, potentially based on collective reference probes.

4.7 Status and relation to the central research question

The shared-memory calculations provide finite-chain diagnostics for a model in which memory is reused and inter-site correlations are retained. They do not yet establish a transfer-matrix representation, a uniform contraction theorem, or a counterexample to such a theorem.

The central question remains whether local reference-stable forgetting can be composed into a spatial attenuation principle for coupled system–memory networks. The shared-memory model is intended to test this question under conditions that are not captured by the fresh-child branching model. Any eventual conclusion must be based on a validated transfer construction or an independent rigorous argument.

5 Conclusion and Future Directions

This work investigates whether local information loss in finite-dimensional quantum memories can provide a mechanism for restricting the propagation of correlations in spatially extended quantum dynamics. The central question is whether local reference-stable forgetting can be composed into a general spatial attenuation bound for coupled system–memory networks. The results developed here establish a concrete local mechanism and examine its compositional prospects in a branching model, while identifying the additional mathematical requirements for a network-level theorem.

5.1 Established local contraction

The first result establishes reference-stable contraction for repeated partial-SWAP collisions between a system qubit and a memory qubit subject to depolarization. Under the stated assumptions, the two-collision block contracts the trace distance between joint system–reference states with a common reference marginal by a factor bounded above by $1 - \epsilon$, where

$$\epsilon = (1 - \lambda)^2 \sin^2(2\theta).$$

Iteration yields the corresponding bound over successive collision blocks. The estimate is uniform with respect to the external reference and therefore controls correlations with degrees of freedom beyond the local system.

This result is a local dynamical statement. It establishes a rigorous example of reference-stable forgetting, but does not by itself imply attenuation with spatial distance, branching depth, or the size of a coupled network.

5.2 Branching attenuation: status and limitations

The second result studies a specific branching-channel construction based on partial-SWAP interactions and child depolarization. Its transfer-tensor formulation provides a framework for analyzing the recursive propagation of operator components through branching generations. The analysis also makes explicit why the local contraction theorem cannot simply be iterated: correlations generated at one stage enlarge and modify the effective reference for later stages, and the required reference-marginal condition is not automatically preserved.

The branching attenuation theorem must therefore be read according to its stated verification status. The transfer-tensor calculations and recursive estimates provide a model-specific route toward attenuation, but the finite-depth base-case certificate and the end-to-end verification remain outstanding. Accordingly, the corresponding induction and attenuation conclusion are conditional on the stated hypotheses and verification obligations. They are not presented here as an unconditional theorem, nor as a consequence of the first result alone.

The shared-memory extension raises a further, distinct issue. In a model where the same memory is reused across sites and successive sweeps, correlations can accumulate through the common memory and across the chain. Finite-chain transfer representations and numerical probe statistics can provide diagnostics, but they do not establish a uniform-in-system-size contraction bound. The conjectured behavior and numerical observations in this direction remain exploratory until the transfer representation, boundary contractions, and relevant norm estimates are rigorously certified.

5.3 Toward a general spatial attenuation principle

A general spatial attenuation theorem requires a compositional mechanism that remains valid in the presence of correlations generated by the dynamics. Several steps appear necessary.

First, one must formulate a local contraction estimate on a precisely specified constrained operator space, with arbitrary external references and with the reference-marginal conditions stated in a form stable under the intended composition. Where such stability is absent, the evolution of the relevant correlations must be tracked explicitly rather than discarded.

Second, one must derive a recursive description of the coupled network dynamics that includes shared-memory and boundary effects. For branching systems, this may take the form of a transfer-tensor recursion with a rigorously controlled invariant subspace. For reused-memory chains, it requires accounting for the connection between the end of one sweep and the beginning of the next, including correlations retained by the memory.

Third, the finite-depth estimates must be converted into bounds uniform in network size. This requires rigorous control of the base cases, the contraction constants, and any prefactors arising from the transfer representation. In particular, a spectral gap alone may not provide the required uniform trace-norm or reference-stable estimate.

Finally, one must relate the resulting attenuation to spatial geometry. A successful general principle should identify the assumptions under which local forgetting yields decay with graph distance, branching depth, or another appropriate measure of separation. Such a result would clarify when memory dynamics impose restrictions beyond those supplied by ordinary locality and finite propagation speed.

The present work thus establishes a local reference-stable contraction mechanism and develops a specific branching model for investigating its composition. The broader spatial attenuation principle remains open. Resolving it will require not merely iterating local contraction factors, but proving that the relevant contraction survives the correlations, shared resources, and boundary effects generated by the network itself.

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